

MICROVISION

Service Centre Flow & WhatsApp Integration

Theory Document v1.0 — Full SC Lifecycle, Status Flows, WhatsApp Triggers, Billing Logic

Admin Assign → SC Portal → Done / Not Done / Part Pending / Replacement → Billing

1. Overview

This document describes the complete lifecycle of a complaint from the moment an admin assigns it to a Service Centre (SC), through all possible paths (Done, Not Done, Part Pending, Replacement), to the final billing step. It also documents every WhatsApp message trigger across the entire flow.

Key principles that govern this entire flow:

- Bill is generated **ONLY** when the complaint reaches a final Done status and admin confirms it. Never before.
- A complaint can cycle through Not Done and Part Pending multiple times before reaching Done. Each cycle is recorded in the complaint timeline.
- Done can only be confirmed once — it is the permanent final closure of a complaint.
- Replacement is a sub-case of Part Pending — the admin delivers a full replacement unit instead of a single part. The flow is identical.
- WhatsApp messages are sent to SC for assignment, reminders, and part/replacement delivery. WhatsApp is sent to the customer only once — when SC accepts the complaint.
- SC fills all billing details (petrol, extras, visits) at the time of marking Done. Admin reviews, can edit, then confirms. Bill locks permanently on admin confirmation.

2. Admin Assigns Complaint to SC

Admin registers the complaint and assigns it to a Service Centre. This is the starting point of the SC flow.

WhatsApp Trigger WA-01 — Sent to SC immediately on assignment OR reopen

- Recipient: SC's registered WhatsApp number
- Timing: Immediately when admin clicks Assign
- Content: Complaint ID, Customer name, Phone, Full address (local address + city + district), Product type (LED/Cooler), Complaint type (Installation/Complaint), Warranty status (In Warranty / Out of Warranty), Admin notes (if any), Portal login URL, REOPENED flag + original Complaint ID if this is a reopen

2.1 No Action Reminder System — SC has not accepted or rejected

Once assigned, if the SC takes no action the system sends automated WhatsApp reminders:

Time Since Assignment	Trigger	Recipient
24 hours — no action	WA-02: Reminder — you have a new complaint pending your response	SC
48 hours — still no action	WA-03: Reminder (repeat)	SC
Every 2 days after that	WA-0X: Repeat reminder every 2 days until SC acts	SC

Reminder cycle stops the moment SC either Accepts or Rejects. There is no auto-rejection or auto-escalation — admin must manually reassign if SC keeps ignoring.

3. SC Rejects the Complaint

SC taps Reject in their portal. Complaint status moves to rejected_by_sc.

- Admin sees this in the Action Centre immediately (badge count updates).
- Admin reviews and reassigns to a different SC (or the same SC again — no block).
- On reassignment, the full cycle from Section 2 restarts — WA-01 fires again to the newly assigned SC.
- The rejection is recorded in the complaint timeline (who rejected, when, any rejection note).

4. SC Accepts the Complaint

SC taps Accept. Complaint status moves to accepted.

WhatsApp Trigger WA-04 — Sent to Customer immediately on SC acceptance

- Recipient: Customer's phone1 (and phone2 if exists)
- Timing: Immediately when SC taps Accept
- Content: Complaint ID (customer's reference number), Product type (LED/Cooler), Complaint type (Installation/Complaint), SC business name, SC phone number, Acknowledgment message that SC has accepted and will visit soon

4.1 No Action After Acceptance — Reminder System

Once SC accepts the complaint, if they take no further action (do not mark Going, Done, Not Done, or Part Pending), the system sends automated WhatsApp reminders:

Time Since Acceptance	Trigger	Recipient
24 hours — no action after acceptance	WA-04B: Reminder — you have accepted a complaint but have not updated its status. Please visit the customer and update.	SC
Every 2 days after that	WA-0X: Repeat every 2 days until SC takes any action (Going, Done, Not Done, or Part Pending)	SC

Reminder cycle stops the moment SC takes any action on the complaint — marks Going, Done, Not Done, or Part Pending. Going alone counts as an action and resets/stops the timer.

4.2 SC Marks Going (Optional)

- SC can optionally tap Mark as Going when they are travelling to the customer.
- This is informational only — no WhatsApp trigger, no form, just a status update.
- Status moves to going. No blocking — SC can skip this entirely and go straight to a conclusion status.

5. SC Marks Done

SC visited the customer, completed the work, and is marking the complaint as resolved.

5.1 Done Form — Fields

Form slides open when SC taps Mark as Done

Field	Required?	Details
Photos	Min 1, max 5	Proof of work completed — site photos, installation proof, repair evidence
Audio note	Optional	Max 2 minutes. SC's verbal description of work done.
Text notes	Optional	Additional written notes about the visit
Petrol / travel charges	Optional (recommended)	Amount field + distance traveled (km) + number of visits total. Subject to 3-lock system (see Section 5.2).
Extra charges	Optional	Multiple line items allowed. Each has: description/reason + amount. SC can add as many as needed.

5.2 Petrol 3-Lock System (unchanged from GRD)

- Edit 1 — Admin sets estimate at complaint registration time.
- Edit 2 — SC fills actual petrol amount + distance + visits in the Done form. This is edit 2.
- Edit 3 — Admin can override the SC's amount if too high. This is edit 3.
- After edit 3, the petrol field is permanently locked. All 3 values stored separately for audit.

For complaints that went through multiple Not Done cycles before Done: SC fills OVERALL totals in this single Done form — total petrol for ALL visits combined, total distance across all visits, total visit count. System does not ask per-visit separately.

5.3 After SC Submits Done Form

- Complaint status moves to done (pending admin confirmation).
- Admin sees this in the Action Centre.
- No WhatsApp is sent to anyone at this stage.

5.4 Admin Reviews and Confirms Done

Admin opens the complaint from the Action Centre. Before confirming, admin can:

- Edit petrol amount (Edit 3 — if SC's amount seems too high). This locks the petrol field permanently.
- Approve or reject individual extra charge line items submitted by SC.
- Add admin-side extra charges if needed.
- Add admin notes.
- Edit any other complaint detail before locking.

Once satisfied, admin clicks Confirm Done. This action:

- Moves complaint status to closed (permanent — cannot be changed after this).
- Generates and locks the bill based on: Preset price (in-warranty only) + Petrol final + All approved extra charges.
- Bill is now visible in the Billing tab for both admin and SC.
- No WhatsApp is sent after Done confirmation.

If admin disputes the SC's work — admin clicks Dispute instead of Confirm. Complaint reverts to accepted. SC must re-submit. A note is attached explaining why it was disputed.

6. SC Marks Not Done

SC visited the customer but could not complete the work (customer not home, access issue, scope mismatch, etc.). This is NOT a closure — the complaint remains open and SC is expected to return.

6.1 Not Done Form — Fields

Form slides open when SC taps Mark as Not Done

Field	Required?	Details
Reason text box	One of these two is compulsory	Written explanation of why work could not be completed
Voice note	One of these two is compulsory	Max 2 minutes. At least one of reason text OR voice note must be provided.
Photos	Optional	Max 5 photos. Not compulsory for Not Done — SC may not have anything to photograph.

Either the reason text box OR the voice note must be filled. Both can be filled. Neither can be empty — system blocks submission if both are blank.

6.2 After SC Submits Not Done

- Complaint status stays as not_done (complaint remains open — not closed).
- Recorded in complaint timeline: who marked not done, when, reason.
- Admin sees this in Action Centre for awareness — no required action from admin at this point.
- No WhatsApp sent immediately on Not Done submission.

6.3 Not Done Reminder System — SC must eventually act

Time Since Not Done Submission	Trigger	Recipient
24 hours — no further action by SC	WA-05: Reminder — complaint still open, please update status	SC
Every 2 days after that	WA-0X: Repeat reminder every 2 days until SC acts	SC

Reminder cycle stops when SC takes any action: marks Done, Part Pending, Replacement, or another Not Done submission. Another Not Done resets the 24-hour timer.

6.4 SC Returns and Acts Again

SC can come back at any time and choose any conclusion:

- Mark Done — triggers the full Done form (Section 5). Petrol and extras in this form cover ALL visits (total across entire complaint lifecycle).
- Mark Part Pending — triggers Part Pending flow (Section 7).
- Mark Not Done again — triggers Not Done form again, timeline records another not_done entry, reminder cycle resets.

7. SC Marks Part Pending (including Replacement)

SC visited, diagnosed the issue, but cannot complete the work because a part (or the entire unit) needs to be sourced and delivered. Replacement is a sub-case of Part Pending where the full product unit is delivered instead of a single component.

7.1 Part Pending Form — Fields

Form slides open when SC taps Mark as Part Pending

Field	Required?	Details
Voice note	Compulsory	Max 2 minutes. SC explains what part is needed and why.
Text notes	Compulsory	Written description of the issue and part requirement.
Parts detail	Compulsory	Name/description of part(s) needed. If replacement: clearly state full unit replacement required.
Photos	Min 2 compulsory, max 5	Proof of diagnosis — damaged part, fault evidence, unit condition.

7.2 After SC Submits Part Pending

- Complaint status moves to part_pending.
- Admin sees this in Action Centre.
- Admin reviews the request — understands what part or unit is needed.
- Admin arranges the part/replacement unit (sourcing, ordering — happens outside the app).
- When part/unit is ready and dispatched, admin marks Delivered in the app.
- This Delivered marker appears as a timeline entry on the complaint — date and admin note (e.g. 'Part dispatched via courier, tracking: XXXX' or 'Replacement unit dispatched').
- No form required from admin for this step — just a timeline marker with an optional note.

WhatsApp Trigger WA-06 — Sent to SC when Admin marks Part/Unit as Delivered

- Recipient: SC's WhatsApp number
- Timing: Immediately when admin clicks Mark as Delivered
- Content: Complaint ID, Customer name, Address, Part/unit delivery notification, Admin note (if any)

7.3 SC Marks Part / Unit Received

- SC receives the part or replacement unit in real life.
- SC opens the complaint in their portal and taps Mark as Received.

- This adds a Received marker to the complaint timeline.
- Complaint status moves to part_received (or remains part_pending — implementation detail, but timeline clearly shows received state).

7.4 No Action After Received — Reminder System

Time Since SC Marked Received	Trigger	Recipient
24 hours — no further action	WA-07: Reminder — part received, please complete the job and update status	SC
Every 2 days after that	WA-0X: Repeat every 2 days until SC acts	SC

7.5 SC Acts After Receiving Part / Unit

SC goes to the customer and completes (or attempts) the work. SC then chooses:

- Mark Done — full Done form (Section 5). Overall billing covering entire complaint lifecycle submitted here.
- Mark Not Done — Not Done form (Section 6). Complaint stays open, 24-hour reminder cycle begins.
- Mark Part Pending again — a new Part Pending cycle starts. Another part is needed. Full Part Pending form again. Admin delivers again. Cycle repeats.

Replacement is handled exactly like Part Pending — the only difference is that the 'part' being delivered is the full product unit. The admin marks Delivered the same way. SC marks Received the same way. Same reminders, same timeline entries. No separate flow.

8. Billing — When and How

Scenario	When Bill is Generated
Normal Done (first visit)	Admin confirms Done → bill generated and locked immediately
Done after multiple Not Done cycles	Admin confirms the FINAL Done → single bill generated covering all visits (petrol + extras SC filled in Done form are overall totals)
Done after Part Pending cycle(s)	Admin confirms Done → single bill generated. SC fills overall totals in final Done form.
Done after Replacement (full unit delivered)	Same as Part Pending — admin confirms Done, single bill generated
Not Done — never reaches Done	No bill ever generated. Complaint stays open indefinitely.
Part Pending — never reaches Done	No bill generated until Done is eventually confirmed.

8.1 Bill Components

Component	In Warranty	Out of Warranty
Preset base price	Included — locked at complaint creation	Not applicable
Petrol / travel	SC fills overall total in Done form (edit 2). Admin can override (edit 3).	Not applicable — Microvison does not cover petrol for out-of-warranty
Extra charges (SC-requested)	Each line item approved/rejected by admin before billing	Same — admin approves/rejects
Extra charges (admin-added)	Auto-approved	Auto-approved — only Microvison-side extras
Customer payment	Not collected by SC (Microvison pays SC)	SC collects directly from customer. SC fills 'Amount collected from customer' field. Record only — never in Microvison invoice.
Total (Microvison owes SC)	Preset + petrol + approved extras	Approved admin extras only (if any). Zero if none.

9. Complete WhatsApp Trigger Summary

Trigger ID	Event	Recipient	Timing
WA-01	Admin assigns complaint to SC (new or reopened)	SC	Immediately on assignment or reopen-assignment. If reopen: includes REOPENED flag + original complaint ID
WA-02	SC has not acted for 24 hours after assignment	SC	24 hours after WA-01 if no action
WA-03	SC still not acted after 48 hours	SC	48 hours after WA-01 if no action
WA-0X	SC still not acted — ongoing	SC	Every 2 days after WA-03 until SC acts
WA-04	SC accepts the complaint	Customer (phone1 + phone2)	Immediately on SC acceptance
WA-04B	SC accepted but no action for 24 hours (no Going, Done, Not Done, or Part Pending)	SC	24 hours after acceptance if no action
WA-0X	SC still not acted after acceptance	SC	Every 2 days after WA-04B until SC acts
WA-05	SC submitted Not Done, no further action for 24 hours	SC	24 hours after Not Done submission
WA-0X	SC still not acted after Not Done	SC	Every 2 days after WA-05 until SC acts
WA-06	Admin marks Part / Replacement unit as Delivered	SC	Immediately when admin marks Delivered
WA-07	SC marked Received but no further action for 24 hours	SC	24 hours after SC marks Received
WA-0X	SC still not acted after Received	SC	Every 2 days after WA-07 until SC acts

No WhatsApp is sent to customer after SC acceptance. No WhatsApp on Done, Not Done, Part Pending submissions, or on final bill generation. No reminder if admin marks Delivered but SC has not marked Received. Customer communication is limited to WA-04 only.

10. Complete Status Flow

Status	Set By	Meaning	Next Possible Statuses
new	System	Complaint created, not yet assigned	assigned
assigned	Admin	Assigned to SC, awaiting SC response	accepted, rejected_by_sc
rejected_by_sc	SC	SC rejected — admin must reassign	assigned (admin reassigns)
accepted	SC	SC accepted — will visit customer	going, done, not_done, part_pending
going	SC	SC is on the way (optional)	done, not_done, part_pending
done	SC	SC completed work — pending admin confirmation	closed (admin confirms), accepted (admin disputes)
not_done	SC	SC could not complete — complaint stays open	done, not_done, part_pending
part_pending	SC	Part or unit needed — awaiting admin delivery	part_received (after admin marks delivered + SC marks received)
part_received	SC	Part/unit received — SC will act	done, not_done, part_pending
closed	Admin	Final — admin confirmed Done, bill generated	reopened (if eligible within 30 days)
reopened	Admin	New complaint created referencing this one	Full new complaint lifecycle

11. Complaint Timeline — What Gets Recorded

Every action in the complaint lifecycle is recorded as a timeline entry. This is the full audit trail visible on the complaint detail view.

Event	Recorded In Timeline
Complaint created	Date, time, admin who created, customer + product details
Assigned to SC	Date, time, which SC, admin who assigned
SC rejected	Date, time, SC name, rejection note (if any)
Reassigned to new SC	Date, time, new SC name, admin who reassigned
SC accepted	Date, time, SC name
SC marked Going	Date, time
SC marked Done	Date, time, photos, audio, notes, petrol, extra charges submitted
Admin disputed Done	Date, time, admin note explaining reason
SC marked Not Done	Date, time, reason text/audio, photos
SC marked Part Pending	Date, time, voice note, notes, parts detail, photos
Admin marked Part/Unit Delivered	Date, time, admin note
SC marked Part/Unit Received	Date, time
Admin confirmed Done (Closed)	Date, time, final bill amount
Reopened	Date, time, reopen notes, reference to new complaint ID

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